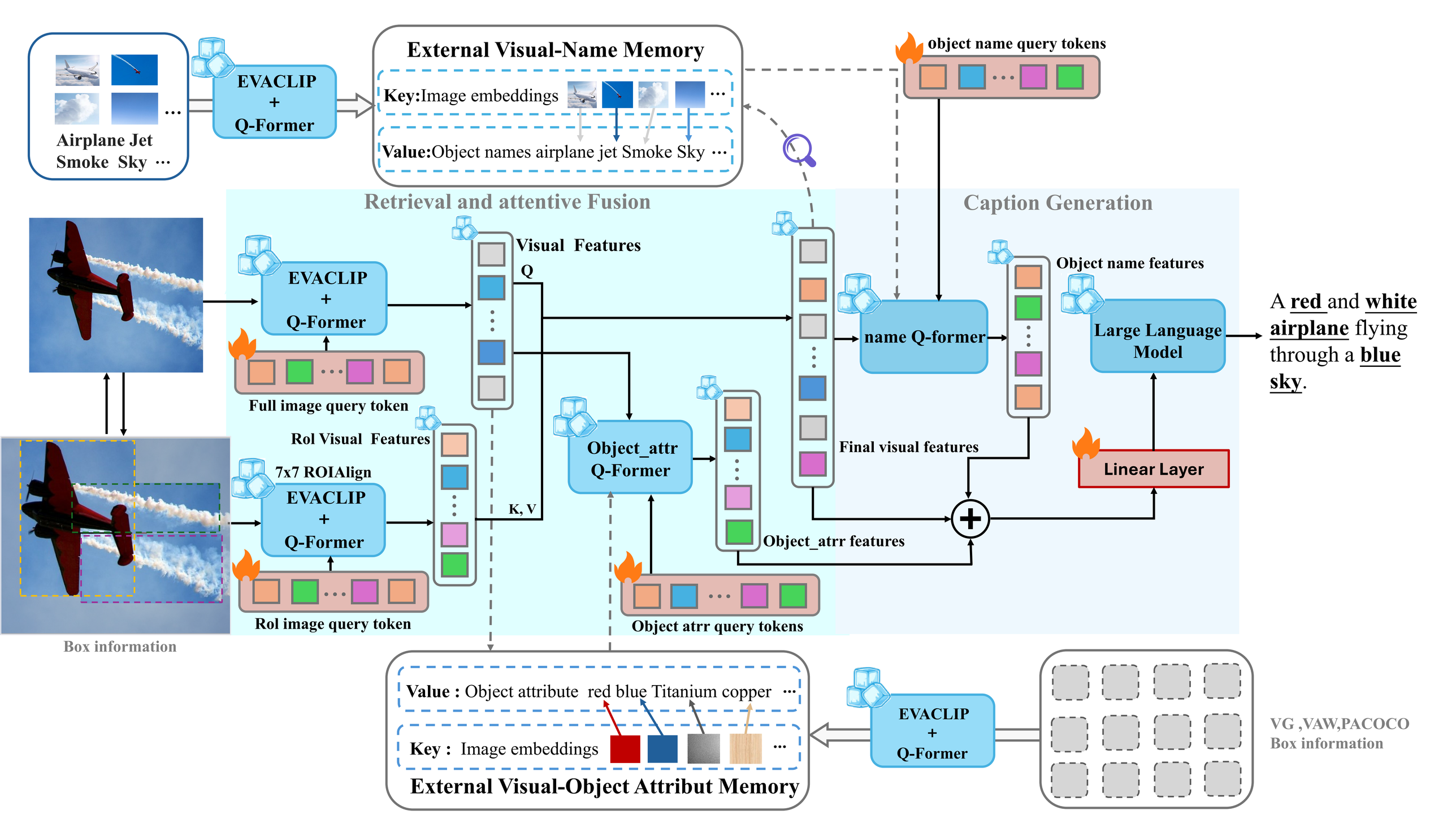

Airplane Jet
Smoke Sky ···
EVACLIP + Q-Former
External Visual-Name Memory
Key:Image embeddings ···
Value:Object names airplane jet Smoke Sky ···
object name query tokens
Retrieval and attentive Fusion
Caption Generation
Visual Features
Q
EVACLIP + Q-Former
Full image query token
Rol Visual Features
7x7 ROIAlign
EVACLIP + Q-Former
Rol image query token
K, V
Object_attr Q-Former
Object atrr query tokens
Object_atrr features
Final visual features
name Q-former
Object name features
Large Language Model
Linear Layer
A red and white airplane flying through a blue sky.
Box information
Value : Object attribute red blue Titanium copper ···
Key : Image embeddings ···
External Visual-Object Attribut Memory
EVACLIP + Q-Former
VG ,VAW,PACOCO
Box information